Revision · Ratio, proportion and rates

Cambridge insert: sharing, scaling and comparing — the quiet workhorse of every exam paper.

LESSON 97–98 | 2 × 40 MIN | ALGEBRA 8 — REVISION | STAGE 9 · 11.1–11.2

LESSON CLOCK

6' | 10' | 10' | 10' | 4'

Simplify five ratios	6 min
Sharing in a ratio	10 min
Direct proportion	10 min
Inverse proportion and rates	10 min
Homework	4 min

LEARNING OBJECTIVES

- Simplify a ratio and share a quantity in a given ratio.
- Use direct proportion, including the unitary method.
- Recognise and use inverse proportion.
- Work with rates, including speed and best value.

TEXTBOOK

National	Annual revision
Cambridge	Learner's Book pp. 238–252
Workbook	Workbook 11.1–11.2

01 Explanation

Ratio and sharing

A ratio compares parts. Simplify it exactly as you would a fraction, by dividing every term by the same number:

$$18 : 24 : 30 = 3 : 4 : 5$$

SHARING IN A RATIO

1. Add the parts — that is how many equal shares there are.
2. Divide the total by the number of parts to get **one part**.
3. Multiply back up for each share.

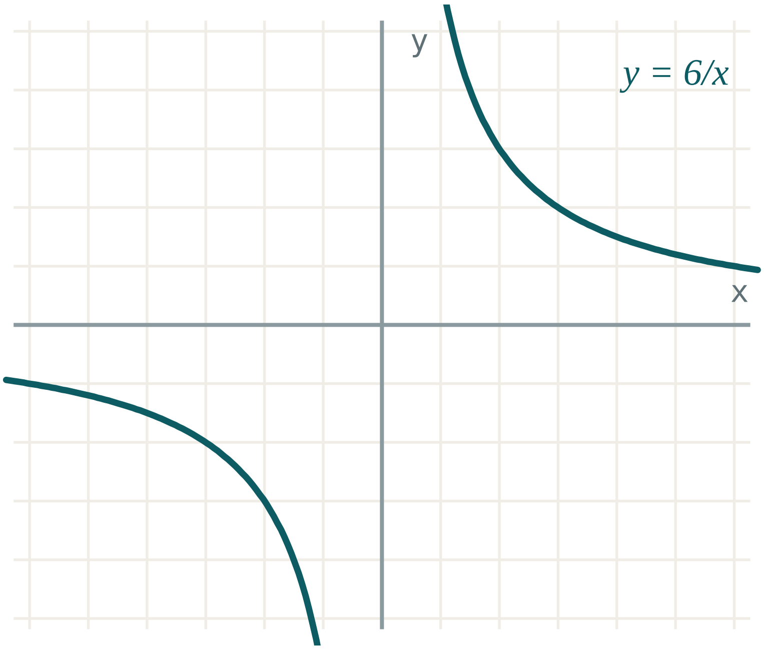
Share $600\ 000$ so'm in the ratio $2 : 3 : 5$: there are 10 parts, so one part is $60\ 000$, giving $120\ 000$, $180\ 000$ and $300\ 000$. Check the sum — it must return the original total.

RATIO ≠ FRACTION

In $2 : 3$ the first quantity is $\frac{2}{5}$ of the total, not $\frac{2}{3}$. $\frac{2}{3}$ is how it compares with the *other part*.

Direct and inverse proportion

	DIRECT	INVERSE
In words	double one, double the other	double one, halve the other
Rule	$y = kx$	$xy = k$
Graph	straight line through $(0, 0)$	a hyperbola
Example	cost of petrol vs litres	workers vs days to finish



Inverse proportion: as one quantity grows the other shrinks, and their product stays the same.

The **unitary method** settles almost every direct-proportion question: find the value of one, then multiply. If 7 books cost $84\,000$ so‘m, one costs $12\,000$, so 11 cost $132\,000$.

For inverse proportion, find the **product** instead. If 4 workers need 18 days, the job is $4 \cdot 18 = 72$ worker-days; 6 workers need $72 \div 6 = 12$ days.

Rates and best value

A **rate** compares two different units — km/h, so‘m per kg, litres per 100 km.

$$speed = \frac{\text{distance}}{\text{time}}, \quad distance = speed \cdot time, \quad time = \frac{\text{distance}}{\text{speed}}$$

For **best value**, reduce every offer to the same unit and compare:

PACK	PRICE	PRICE PER KG
2 kg	*36 000*	*18 000*
3 kg	*51 000*	*17 000*
5 kg	*87 500*	*17 500*

The 3 kg pack is the best value — bigger is not automatically cheaper, which is exactly what the question is testing.

UNITS FIRST

45 minutes is 0.75 hours, not 0.45. Convert before dividing, or the speed is nonsense.

02 Worked examples

EXAMPLE 1 **Share 84 sweets between A, B and C in the ratio $3 : 4 : 5$.**

1. $3 + 4 + 5 = 12$

   Twelve equal parts.

2. $84 \div 12 = 7$

   One part.

3. $A = 21, B = 28, C = 35$

   Multiply back up.

4. $21 + 28 + 35 = 84$

   Check ✓

ANSWER

$21, 28, 35$

EXAMPLE 2 **5 workers build a wall in 12 days. How long would 8 workers take?**

1. More workers, fewer days — **inverse** proportion.
   Do not multiply by $\frac{8}{5}$.

2. $5 \cdot 12 = 60$ worker-days
   The size of the job.

3. $60 \div 8 = 7.5$

4. 7½ days.
   Fewer than 12, as expected.

ANSWER

$7.5\ days$

EXAMPLE 3 **A train covers 210 km in 2 h 30 min. Find its average speed, and how far it goes in 40 minutes.**

1. $2\ h\ 30\ min = 2.5\ h$
   Convert first.

2. $v = \frac{210}{2.5} = 84\ km/h$

3. $40\ min = \frac{2}{3}\ h$

4. $84 \cdot \frac{2}{3} = 56\ km$

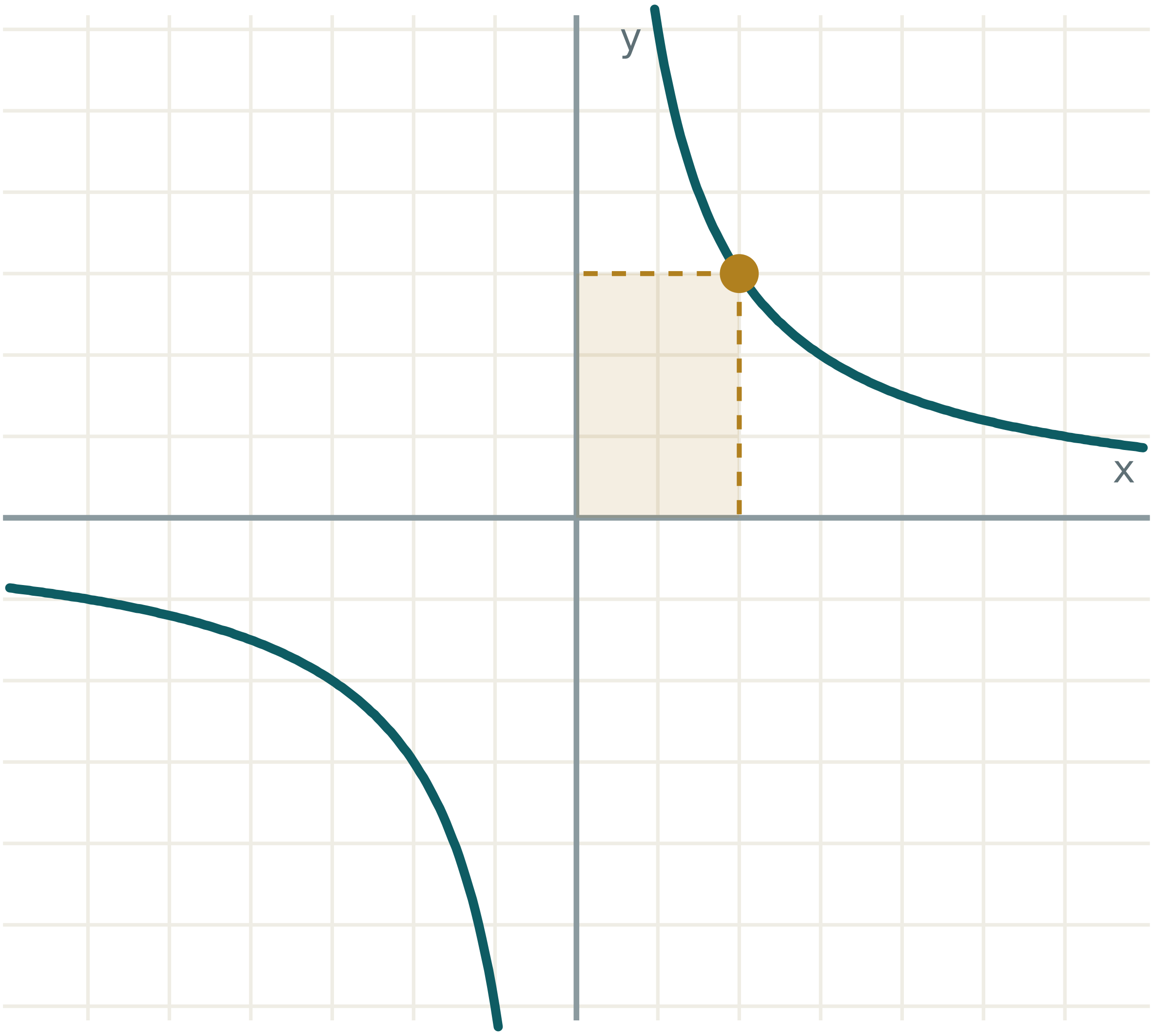
ANSWER

$84\ km/h;\ 56\ km$

03 Interactive model

Slide the value and watch the product stay constant — that is what inverse proportion means.

The graph of $y = k/x$



k **6** | x **2** | y = k/x **3** | x · y **6** | quadrants **I and III**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Ratio	Nisbat	Отношение
Proportion	Proporsiya	Пропорция
Direct proportion	To‘g‘ri proporsionallik	Прямая пропорциональность
Inverse proportion	Teskari proporsionallik	Обратная пропорциональность
Unitary method	Birlikka keltirish usuli	Метод приведения к единице
Scale factor	O‘xshashlik koeffitsienti	Коэффициент подобия
Rate	Tezlik (o‘lchov)	Скорость (величина)
Best value	Eng foydali narx	Наиболее выгодная покупка
Share in a ratio	Nisbatda taqsimlash	Разделить в отношении
Constant of proportionality	Proporsionallik koeffitsienti	Коэффициент пропорциональности

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

12 : 18 in its simplest form is:

- *6 : 9*
- *2 : 3*
- *3 : 2*
- *4 : 6*

90 shared in the ratio *2 : 3* gives:

- *30 and 60*
- *36 and 54*
- *45 and 45*
- *20 and 70*

6 pipes fill a tank in 10 h. Four pipes take:

- *15 h*
- *6.7 h*
- *20 h*
- *8 h*

3 kg costs *45 000*. What do 7 kg cost?

- *90 000*
- *105 000*
- *115 000*
- *135 000*

150 km in 1 h 15 min is a speed of:

- *100 km/h*
- *120 km/h*
- *125 km/h*
- *115 km/h*

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. *Simplify 20 : 35.*

   ANSWER *4 : 7*

2. *Share 60 in the ratio 1 : 2 : 3.*

   ANSWER *10, 20, 30*

3. *4 pens cost 20 000 so‘m. Find the cost of 7.*

   ANSWER *35 000 so‘m*

4. *A car travels 180 km in 3 h. Find its speed.*

   ANSWER *60 km/h*

5. *Simplify 0.5 : 2.*

   ANSWER *1 : 4*

6. *If $y = 4x$, find y when $x = 7$.*

   ANSWER *28*

7. *2 kg of rice costs 24 000. Find the price per kg.*

   ANSWER *12 000 so‘m*

Medium · the standard question

7 PROBLEMS

1. *Share 720 000 so‘m in the ratio 4 : 5 : 7.*

   ANSWER *180 000, 225 000, 315 000*

2. *9 workers finish in 20 days. How long for 12 workers?*

   ANSWER *15 days*

3. *5 books cost 87 500. Find the cost of 8.*

   ANSWER *140 000 so‘m*

4. *A journey of 240 km takes 3 h 20 min. Find the speed.*

   ANSWER *72 km/h*

5. *y is directly proportional to x, and $y = 18$ when $x = 6$. Find y when $x = 11$.*

   ANSWER $k = 3, y = 33$

6. *Which is better value: 4 kg for 62 000 or 6 kg for 90 000?*

   ANSWER *6 kg — 15 000/kg* against *15 500/kg*

7. *The ratio of boys to girls is 5 : 4 and there are 20 boys. Find the class size.*

   ANSWER *36*

Hard · stretch and reason

7 PROBLEMS

1. *A and B share money in the ratio 3 : 7. B gets 60 000 more than A. Find the total.*

   ANSWER *150 000 so‘m*

2. *y is inversely proportional to x, and $y = 8$ when $x = 3$. Find y when $x = 12$.*

   ANSWER $k = 24$, $y = 2$

3. *Three people share 500 000 in the ratio 2 : 3 : 5. How much more does the last get than the first?*

   ANSWER *150 000 so‘m*

4. *A car does 90 km at 60 km/h and 90 km at 90 km/h. Find the average speed for the whole trip.*

   ANSWER *72 km/h — 180 ÷ 2.5*, not the mean of 60 and 90

5. *The ratio $a : b$ is $2 : 3$ and $b : c$ is $4 : 5$. Find $a : b : c$.*

   ANSWER $8 : 12 : 15$

6. *6 machines make 480 parts in 4 h. How many parts do 9 machines make in 5 h?*

   ANSWER *900*

7. *Explain why doubling the workers does not always halve the time in practice.*

   ANSWER The model assumes every worker is equally productive and never gets in another’s way; real jobs have tasks that cannot be split and workers who must wait for each other.

07 Homework

Homework — 6 problems

Cambridge Stage 9, Unit 11. Say “direct” or “inverse” before you calculate.

1. *Simplify 45 : 60 : 75.*
2. *Share 480 000 so‘m in the ratio 3 : 5.*
3. *7 kg costs 91 000. Find the cost of 12 kg.*

4. *8 workers take 15 days. How long do 10 workers take?*
5. *A bus covers 156 km in 2 h 36 min. Find its speed.*
6. *Which is better value: 3 for 21 000 or 5 for 34 000?*